

Insertion Preview: Transient Spectral Oscillations at the 300-Trillion Scale

Joint-manuscript working section

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1 Transient spectral oscillations at the 300-trillion scale

We compare the ordinary prime-count races with low-zero truncations of their standard explicit formulas. This calculation has two deliberately limited purposes. First, it tests whether low Dirichlet zeros account quantitatively for the trajectories observed through 3×10^{14} . Second, it tests the specific finite-scale claim that the complex-character interference has subsided by that range. The calculation concerns the unregularized counting functions $\pi(x; N, a)$. It neither proves the asymptotic proposed for the Gaussian-regularized statistic elsewhere in this paper nor transfers an ordering between the two statistics.

1.1 Normalization and spectral truncation

For a reduced residue class $a \pmod{N}$, set

$$E_N(x; a, 1) := \frac{\varphi(N) \log x}{\sqrt{x}} (\pi(x; N, a) - \pi(x; N, 1)). \quad (1)$$

Let

$$C_N(a) := \#\{b \pmod{N} : b^2 \equiv a \pmod{N}\}.$$

Assuming the critical-line form used in the standard GRH explicit formula, we form the finite truncation

$$\tilde{E}_{N,K}(x; a, 1) := C_N(1) - C_N(a) - \sum_{\chi \neq \chi_0} \sum_{j=1}^K 2 \operatorname{Re} \left(\frac{(\overline{\chi(a)} - 1) e^{i\gamma_{\chi,j} \log x}}{\frac{1}{2} + i\gamma_{\chi,j}} \right), \quad (2)$$

where $\gamma_{\chi,j} > 0$ is the j th positive ordinate returned for $L(s, \chi)$. Thus K means the first K positive ordinates for *each* nonprincipal character, rather than the first K modes globally. The coefficient $\overline{\chi(a)} - 1$ is consistent with selecting the ordered difference between the classes a and 1. It should not be confused with the separate character kernel used to define the regularized statistic.

The ordinates were computed for every nonprincipal character modulo

$$N \in \{7, 8, 11, 19, 23\}$$

through height 80. Two runs of `lfunzeros`, with mesh parameters 64 and 96, were required to agree character by character. Every returned ordinate used in the reconstruction was also required to satisfy

$$|L(\tfrac{1}{2} + i\gamma_{\chi,j}, \chi)| < 10^{-28}.$$

The two mesh computations are robustness checks within one implementation, not an independent proof that the zero list is complete. As a separate cross-check, all three first-zero anchors modulo 8 agree with a pre-existing high-precision `mpmath` computation to 10^{-12} .

Table 1: Top-decade reconstruction of the -1 race. The correlation and RMSE use $K = 25$ zeros per nonprincipal character. Rank and leader changes refer to the observed ordinary-count curves over 52 consecutive intervals.

N	correlation	RMSE	rank changes	leader changes	fraction -1 leads
7	0.965	0.515	17	13	0.358
8	0.931	0.626	23	15	0.358
11	0.939	0.919	30	26	0.377
19	0.971	1.745	17	8	0.415
23	0.826	1.728	39	23	0.132

For the exceptional modulus-19 mode discussed below, a second implementation constructs the character directly with Python FLINT and evaluates its Hardy Z -function using Arb ball arithmetic. Opposite, sign-definite endpoint balls give an independent existence certificate for a critical-line zero. This certificate does not assert uniqueness of that zero, completeness below it, or GRH.

1.2 Finite computational result

The observed data consist of 438 logarithmically spaced checkpoints through 3×10^{14} . The “top decade” below is the 53-point subwindow $x \geq 3 \times 10^{13}$.

Computational Proposition 1 (verified finite computation). *For the immutable input dataset identified in the reproducibility statement below, the checked pipeline produces 13,578 reconstruction rows. At $K = 25$, the correlations between $\tilde{E}_{N,25}(x; -1, 1)$ and $E_N(x; -1, 1)$ on the 53-point top-decade window are respectively*

$$0.964909993, \quad 0.930923926, \quad 0.938516494, \quad 0.971455692, \quad 0.826435845$$

for $N = 7, 8, 11, 19, 23$. Across the 52 consecutive top-decade intervals, the observed rank of -1 changes respectively

$$17, \quad 23, \quad 30, \quad 17, \quad 39$$

times. In particular, for every one of these five moduli the class -1 both attains rank one and leaves rank one within the top decade.

This is a computational proposition: it records the output of a specified, hash-checked finite calculation. It is not a theorem about omitted zeros, GRH, or limiting prime-race behavior.

Figure 1 compares the observed curves with the $K = 1, 3, 10, 25$ truncations. For $N = 7, 11, 19$, even a few modes reproduce much of the slow structure. For $N = 23$, one zero per character is inadequate, and the fit improves only when substantially more modes are retained. This behavior makes a single-mode “onset scale” unreliable: distinct characters and ordinates remain active simultaneously.

1.3 The very low odd-character mode modulo 19

The computation identifies a load-bearing correction to a one-mode description of the modulus-19 race. For the primitive character represented by Conrey index 13 modulo 19, of character order 18, the first returned positive ordinate is

$$\gamma_{19,13,1} = 0.0189563990802261\dots, \quad \chi(-1) = -1. \quad (3)$$

Both mesh runs return this ordinate, and direct evaluation at the zero passes the residual threshold above. Independently, Arb gives a sign-change bracket of width 5.94×10^{-84} around $0.018956399080226142994\dots$ and an upper bound below 3.55×10^{-85} for the modulus of the

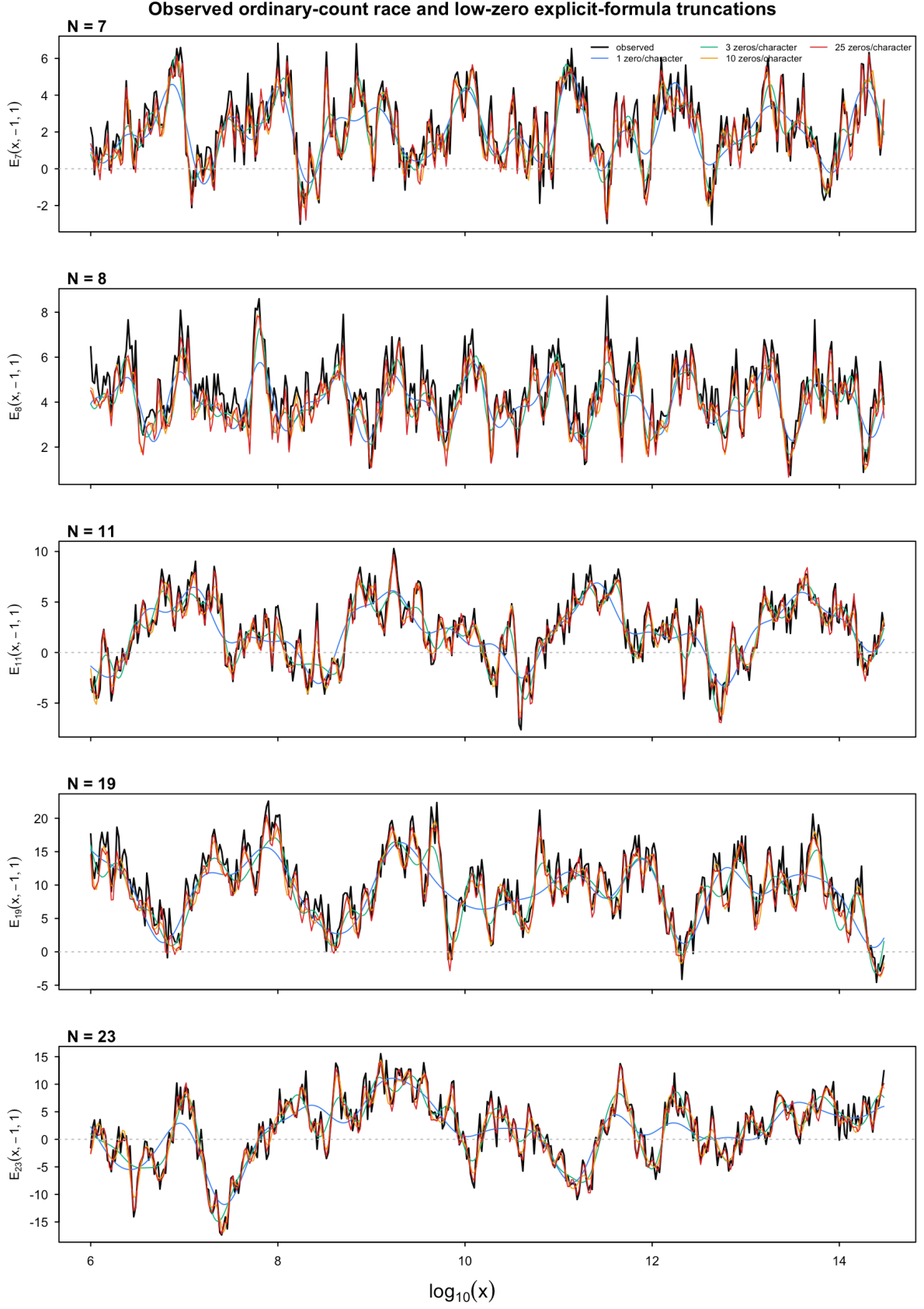


Figure 1: Observed normalized ordinary-count races (black), compared with explicit-formula truncations using the first $K = 1, 3, 10, 25$ positive zeros of every nonprincipal character. The panels show $N = 7, 8, 11, 19, 23$ and the class $a = -1 \pmod{N}$.

midpoint L -value. The intermediate-value theorem therefore certifies existence of at least one zero in the bracket. The corresponding mode is active in the -1 race: on the top decade its root-mean-square contribution is 6.719, and its correlation with the centered observed curve is 0.728.

The conclusion is stable under a substantially deeper truncation. Extending all 17 nonprincipal characters modulo 19 to at least 100 positive ordinates improves the top-decade -1 correlation from 0.9715 at $K = 25$ to 0.9925 at $K = 100$, while the RMSE decreases from 1.745 to 1.252. The $K = 100$ reconstruction retains 14 rank changes and 7 leader changes, with rank and leader agreement against the observed data of 90.6% and 98.1%, respectively. Increasing spectral depth therefore strengthens, rather than removes, the transient-rank conclusion.

Consequently, an estimate based on an ordinate near 1.74 cannot be described as using the lowest relevant complex-character zero modulo 19. The much smaller ordinate in (3) produces a substantially slower oscillation that remains visible near 3×10^{14} . This does not by itself settle the ultimate behavior of the race; it shows that the proposed finite scale is not a demonstrated point beyond complex-character interference.

1.4 Observed rankings and what they do not imply

At the final checkpoint $x = 3 \times 10^{14}$, the class -1 ranks first among the quadratic nonresidues for $N = 7$ and $N = 23$, third of three for $N = 8$, third of five for $N = 11$, and sixth of nine for $N = 19$. Those endpoint ranks are snapshots, and the transition counts in Table 1 show why they cannot be treated as stable asymptotic orderings. The appropriate finite-scale conclusion is:

The unregularized races remain spectrally coherent but dynamically unsettled through 3×10^{14} . Low-zero explicit-formula truncations reproduce the observed trajectories with high correlation, while frequent rank and leader changes persist throughout the top decade. These data support a low-zero transient mechanism, not a universal stabilization threshold or an eventual ordinary-count ordering.

The distinction between statistics is essential. Equations (1)–(2) describe ordinary counts. A theorem about a separately defined Gaussian-regularized spectral statistic would concern that statistic only. Inferring an eventual ordering of ordinary counts would require an additional transfer theorem with explicit error control; no such transfer is supplied by the computation here.

1.5 Reproducibility, data availability, and limitations

The authoritative input is `curve_3e14.tsv`, with SHA-256 digest

57957bdb3ce3243272c3d4b8e9ffe7dfb734b759f48b63becf7ae6f924e1caab.

The accompanying reproducibility package contains the PARI/GP zero generator, the reconstruction program, fit metrics, the complete transition ledger, per-mode attribution, figure source, a negative-test suite for the verifier, and a SHA-256 manifest. Its `independent_n19` follow-up contains the Arb sign-change certificate, the $K = 25, 50, 100$ reconstruction, seven adversarial verifier fixtures, and a separate manifest. Running the two pipeline scripts regenerates their tabular outputs and checks the declared invariants. For an archival release, the package, input dataset, exact software versions, and manifest should be deposited under a permanent DOI; the DOI replaces this sentence before submission.

The numerical interpretation has four limits. First, the reconstruction uses the GRH critical-line form of the explicit formula. Second, finite zero lists do not prove GRH or completeness of the zero set. Third, correlation measures finite explanatory agreement, not mathematical implication. Fourth, neither the observed curves nor their truncations prove a universal or eventual ordering. The results are therefore best used as a controlled finite-scale diagnostic and as a constraint on any proposed asymptotic theory.